

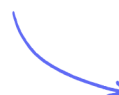
Calculator Questions

Exponentials & Logarithms

Exponential Growth & Decay / Logarithms

Medium (7 questions)	/18
Hard (7 questions)	/29
Very Hard (7 questions)	/23
Total Marks	/70

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Medium Questions

1 $\log(3x) = 3$

Find the value of x .

(2 marks)

- 2 A town has a population of 45 000. This population increases exponentially at a rate of 1.6% per year.

Find the population of the town at the end of 5 years. Give your answer correct to the nearest hundred.

(3 marks)

- 3 The population of a town decreases exponentially at a rate of 1.7% per year.

The population now is 250 000.

Calculate the population at the end of 5 years.

Give your answer correct to the nearest hundred.

(3 marks)

- 4 Newton has a population of 23 000.

The population decreases exponentially at a rate of 1.4% per year.

Calculate the population of Newton after 5 years.

(2 marks)

- 5** There are 30 000 lions in Africa.
The number of lions in Africa decreases exponentially by 2% each year.

Find the number of lions in Africa after 6 years.
Give your answer correct to the nearest hundred.

(2 marks)

- 6** The population of a city is increasing exponentially at a rate of 2% each year.
The population now is 256 000.

Calculate the population after 30 years.
Give your answer correct to the nearest thousand.

(3 marks)

- 7 (a)** Find x .

$$10^x = 350$$

Write your answer to 3 significant figures.

(1 mark)

- (b)** Find y .

$$\log(y) = 2.45$$

Write your answer to 3 significant figures.

(2 marks)

Hard Questions

1 (a) The population of a village increases exponentially at a rate of 6% per year. In 2023 the population was 901.

(i) Calculate the population in 2022.

(ii) Calculate the population in 2026.

(iii) Find the number of complete years it takes for the population of 901 to first become greater than 1600.

(8 marks)

(b) In another village, the population increases exponentially at a rate of $r\%$ per year. At the end of 5 years, the overall increase in the population is 10.41% .

Find the value of r .

(3 marks)

- 2 The value of a gold ring increases exponentially at a rate of 5% per year. The value is now \$882.

(i) Calculate the value of the ring 2 years ago.

\$ [2]

(ii) Find the number of complete years it takes for the ring's value of \$882 to increase to a value greater than \$1100.

[2]

(4 marks)

- 3 Mohsin's earnings increase exponentially at a rate of 8.7% each year.
During 2018 he earned \$195 600.

During 2027, how much **more** does he earn than during 2018?

\$

(3 marks)

- 4 In a city the population is increasing exponentially at a rate of 1.6% per year.
Find the overall percentage increase at the end of 20 years.

..... %

(2 marks)

- 5 Jack bought a new boat for £12 500

The value, £ V , of Jack's boat at the end of n years is given by the formula

$$V = 12500 \times (0.85)^n$$

At the end of how many years was the value of Jack's boat first less than 50% of the value of the boat when it was new?

(3 marks)

- 6 The population of a village is 6400.

The population is decreasing exponentially at a rate of $r\%$ per year.

After 22 years, the population will be 2607.

Find the value of r .

$$r = \dots\dots\dots$$

(3 marks)

- 7 The number of a particular newspaper sold decreases exponentially by $x\%$ each year.

Over a period of 21 years the number of newspapers sold decreases from 1 763 000 to 58 000.

Calculate the value of x .

$$x = \dots\dots\dots$$

(3 marks)

Very Hard Questions

- 1 The population of a village decreases exponentially at a rate of 4% each year.
The population is now 255.

Calculate the population 16 years ago.

(3 marks)

- 2 The population of the Earth in 2017 was estimated to be 7.53×10^9 .
The population of the Earth in 2000 was estimated to be 6.02×10^9 .
Assume that the population of the Earth increased exponentially by $y\%$ each year for these 17 years.

Find the value of y .

$y =$

(3 marks)

- 3 The value of the campsite has increased exponentially by 1.5% every year since it opened 30 years ago.

Calculate the value of the campsite now as a percentage of its value 30 years ago.

..... %

(2 marks)

- 4 The value of Dhanu's model railway is \$550.
This value increases exponentially at a rate of $r\%$ per year.
At the end of 5 years the value will be \$736.

Calculate the value of r .

$r =$

(3 marks)

- 5 A bacteria population increases exponentially at a rate of $r\%$ each day.
After 32 days, the population has increased by 309%.

Find the value of r .

$r =$

(3 marks)

- 6 Over a period of 3 years, a company's sales of biscuits increased from 15.6 million packets to 20.8 million packets.
The sales increased exponentially by the same percentage each year.

Calculate the percentage increase **each year**.

..... %

(3 marks)

7 Tomas has two cars.

(i) The value, today, of one car is \$21 000.

The value of this car **decreases** exponentially by 18% each year.

Calculate the value of this car after 5 years. Give your answer correct to the nearest hundred dollars.

\$ [3]

(ii) The value, today, of the other car is \$15 000.

The value of this car **increases** exponentially by $x\%$ each year. After 12 years the value of the car will be \$42 190.

Calculate the value of x .

$x =$ [3]

(6 marks)